CAROLINE ZIMMER

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Portfolio: carolinezimmer.github.io

EDUCATION

University of Pennsylvania, School of Engineering and Applied Science | Philadelphia, PA
Master of Science in Engineering, Robotics | Accelerated Master's Program, GRASP Lab — Dec 2027
Bachelor of Science in Engineering, Mechanical Engineering | **GPA**: 3.70 / 4.0 — May 2027
Minors: Mathematics, Engineering Entrepreneurship

ETH Zürich, Exchange Semester | Zürich, Switzerland — Spring 2026

Relevant Coursework: Machine Design & Manufacturing, Finite Element Analysis, Design of Mechatronic Systems, Advanced Dynamics, Mechatronics, Engineering Economics

EXPERIENCE

Heliospace Corporation | Mechanical Engineering Intern | San Francisco, CA — Summer 2026

- Produced all SolidWorks models of parts and assemblies of the Quasi-Static Release Mechanism, a NASA SBIR low-shock spacecraft release device combining PEEK and precision metal components
- Led design for manufacturing analysis across the full mechanism, selecting the manufacturing process for each part based on geometries, tolerance requirements, production volume, and cost
- Performed FEA analysis in SolidWorks to verify that applied loads produced deformation required to drive actuation

Enfra Solutions | Mechanical Engineer Intern — Summer 2025

- Delivered finalized AutoCAD models of HVAC ductwork and piping systems for Novant Health hospital construction plans, contributing to the largest Energy-as-a-Service transaction in U.S. healthcare history
- Modeled existing mechanical layouts in Revit, using point cloud data in ReCap to reflect as-built conditions

PROJECTS

Senior Design, Manufacturing Engineer | Tunnel Boring Machine Steering Platform — Fall 2026 – Spring 2027

- Designing a fully electromechanical steering module to replace the hydraulic power pack on a tunnel boring machine, delivering $\pm 5^\circ$ independent yaw and pitch using off-the-shelf linear actuators
- Leading manufacturing planning and sourcing for the articulation joint, selecting stock, process, and vendor for each structural component against a $3.25k build budget

Mechanical Design and Precision Manufacturing | Gamma-type Stirling Engine — Fall 2025

- Designed and modeled a functional Gamma-type Stirling engine assembly in SolidWorks and manufactured every component in-house
- Programmed CNC toolpaths in Mastercam and conducted high-precision machining operations (lathe, mill, threading, reaming) to manufacture components within specified tolerances according to engineering drawings
- Applied GD&T callouts and tolerance stack-up analysis across critical interfaces to improve manufacturability, reliability, and ease of assembly

Mechatronic Systems & Controls | BLDC Motor Balancing System — Fall 2025

- Derived nonlinear equations of motion and a linearized state-space model for a BLDC motor-actuated rotating rod, then designed and tuned a PID controller in MATLAB across perturbation and noise conditions
- Deployed the controller on hardware, commanding throttle via IMU feedback and validating closed-loop response against simulation

LEADERSHIP AND ACTIVITIES

Kappa Alpha Theta Sorority, UPenn Chapter | Academic Development Director — Spring 2025 – Fall 2025
Society of Hispanic Professional Engineers | Member — Fall 2023 – Present

TECHNICAL SKILLS

Programming: C, Python, MATLAB, Java, Arduino

Software/Tools: SolidWorks, Inventor, Mastercam, AutoCAD, Revit, COMSOL, Microsoft Office